# Use Case Template

## CardioAI - Heart Disease Prediction System

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| **1. Use Case Title** | Compute XAI Agreement Metrics |
| **2. Abbreviated Title** | XAI Agreement |
| **3. Use Case Id** | UC-18 |
| **4. Actors** | ML Engine (Supporting) |
| **5. Description** | The system computes agreement and consistency metrics across the three XAI methods (Grad-CAM, SHAP, Attention) to quantify how much they agree on which clinical features are most important, boosting explanation reliability. |
| **5.1. Pre Conditions** | 1. All three XAI outputs (UC-12, UC-13, UC-14) are available.<br>2. Feature importance rankings from each method are normalized. |
| **5.2. Task Sequence** | 1. System normalizes the feature importance rankings from each XAI method.<br>2. System computes pairwise rank correlation (Spearman's rho).<br>3. System computes top-k feature overlap across all 3 methods.<br>4. System computes an overall agreement score.<br>5. System outputs the score and a visual agreement matrix. |
| **5.3. Post Conditions** | 1. An agreement score is computed and displayed on the dashboard.<br>2. Clinician can assess the reliability of XAI explanations. |
| **6. Modification History** | Date: September 11, 2026 |
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